



Alleviation of heavy metal accumulation in rice: A synergistic planetary health approach of organic amendment and phytoremediation in cadmium contaminated soil

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ABSTRACT

Cadmium (Cd) contamination in agricultural soils poses serious risks to crop productivity, food safety, and public health. Therefore, we aimed to mitigate the detrimental effects of Cd stress on plant growth by utilizing organic substrates and phytoremediation methods. In this study, we performed a pot trial to investigate the potential mediating effects of vermicompost (VC) and floating duckweed (DW) on Cd uptake by rice cultivated in Cd-contaminated soil. We applied five distinct levels of VC and DW to optimize the rates of VC-DW applications. Our study revealed overall increases in plant height, root length, panicle length, number of filled-grain panicles per plant, 1000-grain weight, and both above- and below-ground biomass in the presence of VC and DW. Furthermore, VC and DW effectively mitigate Cd's toxic effects on plant growth and chlorophyll content, resulting in an increased grain yield. For example, treatment VC_{4.4}DW₃₄₀ caused significant increases in rice grain g pot⁻¹ (30.4 %) and decreases in grain Cd content (80.0 %) compared to the untreated control. We also found that using VC and DW efficiently minimized the production of malondialdehyde by increasing the activities of superoxide dismutase and catalase. The lower Cd content in rice grains in the presence of optimal VC (4.4 %, weight/weight) and DW (187.0 g m⁻²) rates suggests that applying these precise rates of VC and DW for cultivating rice in Cd-polluted soil could be an effective strategy to address this global planetary health problem.

1. Introduction

Cadmium (Cd) is a toxic, nonessential metal that represents one of the most significant global environmental problems due to its non-decomposable, persistent, and widespread contamination from natural and anthropogenic activities (Khan et al., 2022). Particularly industrial and agricultural activities (e.g., mining and smelting operations, usage of phosphate fertilisers, sewage sludge, and contaminated irrigation water) greatly influence soil Cd pollution (Haider et al., 2021). As harmful to the human (organism) body, fauna, macrophytes, and other plant strains, Cd is a potent threat to planetary health and demands urgent, multidisciplinary action (Dutta et al., 2021). Due to its considerable menace correlated with rice production, its adverse impact on worldwide food safety has been investigated extensively.

Rice (*Oryza sativa* L.), a staple food for more than 50 % of the world's population (Mridha et al., 2022), is a good accumulator of Cd due to its farming in flooded soils, which enhances Cd absorption. Consumption of rice contaminated with Cd can cause serious health issues such as kidney disease, osteoporosis, and cardiovascular disease (Xia et al., 2024). To reduce Cd accumulation in rice grain and other plant species, several researchers have employed numerous techniques, including the incorporation of inorganic and organic amendments, water management strategies, genetic breeding, and phytoremediation techniques (Hussain et al., 2021; Nazia et al., 2021; Zveushe et al., 2023).

Vermicompost (VC) and duckweed (DW) are commonly found in Bangladesh and have shown their abilities to decrease the availability and accumulation of heavy metals in rice (Roy et al., 2024a).

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Vermicomposting is a process in which earthworms transform the organic waste into useful fertilizer (Yatoo et al., 2022), which can positively enhance soil fertility and improve plant growth even under heavy metal-contaminated soil. In addition to facilitate the growth of soil microorganisms, VC stabilized metal ions and thus reduced the bioavailability of heavy metal ions in the soil systems (Das et al., 2024).

According to Wang et al. (2023), VC may improve Cd retention by enhancing the surface charge, thereby transforming the available Cd fraction into an organically bound form. In their several studies (Iqbal et al., 2024a, 2024b, 2024c), reported that VC alleviates Cd toxicity and better yields of rice in a Cd-contaminated soil. In our previous study, we also observed that VC reduces arsenic-induced stress and improves the growth and yield of rice (Roy et al., 2024a). In the case of the phytoremediation approach, we considered that duckweed (DW), a common and abundant aquatic plant, has shown significant potential for phytoremediation of heavy metals, including Cd, from polluted water and soil (Zhou et al., 2023). The rapid growth rate and high biomass production of DW make it an efficient bio-accumulator of heavy metals, and its application in soil amendment has been associated with reduced Cd uptake in rice (Wang et al., 2022). Several authors reported DW as a promising candidate for Cd phytoremediation from different sources (Yang et al., 2023, J. 2024; Zhou et al., 2023). Chen et al. (2023) specifically mentioned that DW reduces the availability of Cd in soil and results in a reduction of Cd content in rice grains.

Even though VC and DW have been studied separately for their potential to enhance plant resistance to Cd stress, their synergistic role has hardly been discovered. It is postulated that the combined use of VC and DW, due to their different mechanisms of action, results in a synergistic effect. The VC improves soil physical structure and microbes diversity, enhances the immobilization of Cd and other heavy metals, and reduces its availability for plant absorption (Wang et al., 2021). On the other hand, DW function as bio-accumulators, they absorb the Cd from the soil thus reducing the Cd concentration in the rhizosphere directly (Wang et al., 2022). As a result, the joint use of the two amendments would lead to a more efficient reduction in Cd bioavailability and uptake by the rice plant than the application of either amendment alone. However, so far, little empirical research on their joint effectiveness of VC and DW for Cd accumulation and remediation has been conducted, demonstrating the novelty of this work.

To address this research gap, this work explored the impact of co-application of VC and DW on rice growth and Cd accumulation by the rice grain in Cd-contaminated soil. More precisely, we quantified the effectiveness of the selected amendment strategy by measuring multiple morphological, physiological, and biochemical growth traits of the rice seedlings. Moreover, in this study, we have applied Response Surface Methodology (RSM) to assess the effect of VC and DW modulation on plant growth and Cd uptake and accumulation in Cd-contaminated soil-grown rice. The RSM is a powerful statistical approach to optimizing the interactions of the input variables affecting the response or output variable that has been widely employed in the agricultural and environmental sciences for similar applications (Roy et al., 2020b, 2020a). The results of this study will be important to apply organic amendments and phytoremediation in limiting Cd accumulation in crops such as rice grains to secure food security and help maintain planetary health.

2. Methodology

2.1. Pot experiment

We carried out a pot experiment in a shelter house at the Sylhet Agricultural University, Bangladesh, from December 2021 to May 2022. We collected soil from the local area (24° 57' 38" N latitude and 91° 53' 54" E longitude) at 0–15 cm depth and stacked it in a net house for air drying. Soil samples were sieved well, and unwanted materials were removed carefully before putting them into the plastic pot. The physical and chemical properties of the experimental soil are presented in

Table S1. Each pot had 5.0 kg of air dried soil, to which five different concentrations of VC (1.0, 1.6, 3.0, 4.4, and 5.0 %, weight/weight) were added.

The soil was flooded with water and soaked for an entire night. The next day, cadmium chloride monohydrate ($\text{CdCl}_2 \cdot \text{H}_2\text{O}$) was solubilized in deionized water and mixed evenly into soil to obtain a final cadmium content of 5 mg kg^{-1} and left for three months to become stabilized. Around 60–70 % field capacity water was maintained during this period. After that, soils were saturated with Cd-free tap water before transplantation of seedlings. We transferred two seedlings, aged forty-five days, into each pot different concentrations of fresh floating DW was then added. A high-yielding hybrid variety of Boro rice, SL 8H, was used because of its common cultivation in the study area.

From the time of transplanting until 7 days before harvesting, Cd-free tap water was kept at a level of about 3–4 cm above the soil in the pots. Growth-promoting applications of urea (120 ppm), triple super phosphate (25 ppm), potassium (40 ppm), sulphur (12.5 ppm), and gypsum (120 ppm) were added to plants. Before planting seedlings, the soil was amended with one-third of the recommended dose of urea and the recommended doses of all other fertilizers. The second and third splits of urea were applied 30 days (at the maximum tillering stage) and 70 days (at the panicle initiation stage) after transplantation, respectively. There were a number of cross-cultural interventions employed during the experimental period.

2.2. Experimental design

Using RSM, we identified the optimum VC and DW application rates to achieve high rice yield and low Cd uptake in rice grains. A CCD (central composite design) was selected to optimize the process. In this investigation, we used VC and DW as independent variables, and rice yield, Cd content, and various morpho-physiological growth parameters as response variables. We chose the VC and DW application rates in our study based on some previous findings (Kiran, 2019; Roy et al., 2024a). Five different levels of VC and DW were chosen (Table 1) based on the following equation:

$$X_i = \frac{Z_i - Z_{i0}}{\Delta Z_i} \quad (1)$$

Here, X_i = coded value of the variables, Z_i = real value variable, Z_{i0} is the real value of Z_i at the center point, and ΔZ_i is the step change value.

According to RSM, in total, twelve treatments (including the control) were developed. Each treatment was replicated three times, and all the pots were randomly arranged in the plastic shade. Table 2 presents the experimental treatments, their numerical codes, and their actual values.

The results of the experiments were evaluated using a second-order polynomial model, as follows:

$$Y = \beta_0 + \beta_1 A + \beta_2 B + \beta_{12} AB + \beta_{11} A^2 + \beta_{22} B^2 \quad (2)$$

Here, Y = response variable, β_0 = constant-coefficient, β_1 and β_2 = linear coefficients, β_{12} = interaction coefficient, β_{11} and β_{22} = quadratic coefficients, and A and B are the coded values of VC and DW, respectively.

2.3. Estimation of morpho-physiological and biochemical growth parameters

Chlorophyll content of rice leaves was determined using a portable Minolta chlorophyll meter (SPAD-502, Osaka 590- 8551, Japan). Rice was harvested when it reached maturity, using scissors made of stainless steel. Whole rice plants were washed at least three times with deionized water and once with ultra-pure water after harvesting, and then they were spread out on a plastic sheet to dry in the sun. Three randomly selected rice plants from each treatment were used to classify yield-related components such as plant height (cm), panicle length (cm),

Table 1
Central composite design (CCD) factor levels.

Independent variables	Codes	Coded and actual values				
		−1.414	−1	0	+1	+1.414
Vermicompost (VC) (% w/w)	A	Lowest/no dose	Low dose	Moderate dose	High dose	Highest dose
Duckweed (DW) (g m ^{−2})	B	1.0	1.6	3.0	4.4	5.0
		0	60	200	340	400

Table 2
Experimental design and treatment combination arrangement in the central composite design.

Treatments	Coded level of factors		Quantity applied	
	Vermicompost (VC)	Duckweed (DW)	VC (% w/w)	DW (g m ^{−2})
1 Control	-	-	0	0
2 VC _{1.6} DW ₆₀	−1	−1	1.6	60
3 VC _{4.4} DW ₆₀	1	−1	4.4	60
4 VC _{1.6} DW ₃₄₀	−1	1	1.6	340
5 VC _{4.4} DW ₃₄₀	1	1	4.4	340
6 VC ₁ DW ₂₀₀	−1.414	0	1.0	200
7 VC ₅ DW ₂₀₀	1.414	0	5.0	200
8 VC ₃ DW ₀	0	−1.414	3.0	0
9 VC ₃ DW ₄₀₀	0	1.414	3.0	400
10 VC ₃ DW ₂₀₀	0	0	3.0	200
11 VC ₃ DW ₂₀₀	0	0	3.0	200
12 VC ₃ DW ₂₀₀	0	0	3.0	200

* Subscripts in the names of treatments denote the rates at which vermicompost and duckweed were applied; for example, VC_{1.6} = Vermicompost @ 1.6 % w/w, and DW₆₀ = Duckweed @ 60 g m^{−2}.

and filled grains per panicle. The plants' roots, panicles, and overall height were all measured using a meter scale. Each panicle's filled and unfilled grains were carefully collected and counted separately. An electric balance was used to record the straw and grains' weights accurately. Both above- and below-ground biomass were measured after being dried in an oven. Using the method [Rahman et al. \(2009\)](#) outlined, rice grains were finely ground and digested. Cd content in rice was determined using an ICP-MS (Elan DRC II, PerkinElmer) instrument. Malondialdehyde (MDA) content was measured in rice leaves using the [Roy et al. \(2022\)](#) methods. Activities of superoxide dismutase (SOD) and catalase (CAT) were determined in accordance with the protocols of [Roy et al. \(2020b\)](#) and [Beers and Sizer \(1952\)](#), respectively.

2.4. Data analysis

Design Expert software (version 11, Stat-Ease, USA) was used to optimize VC and DW rates via Derringer's desirability function approach. Origin 2018 (OriginLab Inc., USA) was used to conduct principal component analysis (PCA) of several growth factors. The ClustVis tool (<https://biit.cs.ut.ee/clustvis/>) was used to create a heatmap.

3. Results

The application of moderate levels of vermicompost (VC) and duckweed (DW) (VC₃DW₂₀₀) resulted in a significant increase in rice plant height when compared to the control (VC₀DW₀) and other treatments (Table 3). Furthermore, the addition of moderate to high levels of VC and DW led to a significant increase in root length, with the greatest enhancement observed when both VC and DW were added together (Table 3). The addition of DW had a greater impact on panicle length compared to VC. Treatments VC_{4.4}DW₃₄₀ and VC₃DW₄₀₀ resulted in a 26.4 % and 24.5 % increase in panicle length, respectively, when compared to the control (Table 3). Additionally, the application of VC and DW across various treatments resulted in a substantial increase in

Table 3
The effects of various regimes of vermicompost-duckweed (VC-DW) on plant height, root length, panicle length, and SPAD value of rice seedlings. Different lowercase letters indicate statistically significant differences at $p < 0.05$. Data are presented as mean $\pm$ standard error ($n = 3$).

Treatments	Plant height cm	Root length cm	Panicle length Cm	SPAD
Control	116.4 $\pm$ 2.76 ^b	34.5 $\pm$ 0.49 ^b	16.3 $\pm$ 0.8 ^d	38.1 $\pm$ 0.77 ^b
VC_{1.6}DW₆₀	124.6 $\pm$ 0.46 ^{ab}	38.65 $\pm$ 0.94 ^{ab}	17.48 $\pm$ 0.68 ^{cd}	44.1 $\pm$ 1.92 ^{ab}
VC_{4.4}DW₆₀	131.8 $\pm$ 0.92 ^{ab}	39.78 $\pm$ 0.66 ^{ab}	18.5 $\pm$ 0.16 ^{abcd}	46.6 $\pm$ 0.27 ^{ab}
VC_{1.6}DW₃₄₀	132.4 $\pm$ 0.86 ^{ab}	39 $\pm$ 0.71 ^{ab}	19.13 $\pm$ 0.46 ^{abc}	45.9 $\pm$ 0.42 ^{ab}
VC_{4.4}DW₃₄₀	133.5 $\pm$ 1.19 ^{ab}	43.15 $\pm$ 0.91 ^a	20.61 $\pm$ 0.33 ^a	49.44 $\pm$ 0.13 ^a
VC₁DW₂₀₀	126.6 $\pm$ 3.52 ^{ab}	38 $\pm$ 1.01 ^{ab}	18.07 $\pm$ 0.32 ^{bcd}	43.57 $\pm$ 0.76 ^{ab}
VC₅DW₂₀₀	134.7 $\pm$ 2.48 ^{ab}	42 $\pm$ 1.03 ^a	19.43 $\pm$ 0.18 ^{abc}	48.14 $\pm$ 1.37 ^a
VC₃DW₀	128.2 $\pm$ 2.62 ^{ab}	38.49 $\pm$ 0.8 ^{ab}	16.29 $\pm$ 0.63 ^d	43.89 $\pm$ 1.19 ^{ab}
VC₃DW₄₀₀	132.8 $\pm$ 2.59 ^{ab}	43.5 $\pm$ 0.69 ^a	20.3 $\pm$ 0.38 ^{ab}	48.47 $\pm$ 0.84 ^a
VC₃DW₂₀₀	140.7 $\pm$ 3.51 ^a	42.83 $\pm$ 1.1 ^a	18.9 $\pm$ 0.15 ^{abc}	47.38 $\pm$ 0.44 ^a
VC₃DW₂₀₀	141.5 $\pm$ 3.48 ^a	43.03 $\pm$ 0.96 ^a	18.6 $\pm$ 0.15 ^{abcd}	47.27 $\pm$ 0.72 ^a
VC₃DW₂₀₀	140.2 $\pm$ 3.16 ^a	43.23 $\pm$ 2.03 ^a	18.76 $\pm$ 0.27 ^{abc}	47.63 $\pm$ 0.77 ^a

the SPAD value of rice leaves, with increases ranging from 14.4 % to 29.8 % when compared to the control (Table 3).

Significant increases in rice grain g pot^{−1} were observed in treatments 3, 5, 7, 10, 11, and 12 with increases of 22.6 %, 30.4 %, 32.1 %, 24.6 %, 26.0 % and 23.4 %, respectively, when compared to the control (Table 4). RSM analysis revealed that although rice grain g pot^{−1} increased with increasing levels of DW and VC, the effect of VC was more prominent than that of DW, as confirmed by the regression coefficient (Table 6). The addition of DW and VC also resulted in a remarkable increase in the number of filled grains and a decrease in the number of unfilled grains (Table 4). Additionally, treatments 5, 7, and 10 significantly increased 1000-grain weight, with increases of 26.0 %, 28.2 %, and 24.1 %, respectively, when compared to the control (Table 4).

All VC–DW treatments markedly enhanced the above- and below-ground biomass compared with that of the control (Table 5). The maximum above-ground biomass (44.52 g) was obtained in the treatment VC₃DW₂₀₀. The maximum value of below-ground biomass was observed in VC₃DW₄₀₀ (26.51 g), and it was found that almost all treatments were significantly greater than that of the control. A significant decrease in grain Cd concentration was found for all amendments in comparison with the control (0.35 mg kg^{−1}). The lowest Cd concentration was discovered in VC_{4.4}DW₃₄₀ (0.07 mgkg^{−1}) followed by VC₅DW₂₀₀ (0.08 mgkg^{−1}), and VC₃DW₄₀₀ (0.09 mgkg^{−1}). The reductions constitute 74–80 % less accumulation of Cd than the control (Table 5). MDA content was highest in control seedlings (2.12 μ molg^{−1} FW), suggesting increased oxidative damage. All VC–DW treatments significantly lowered MDA and VC₃DW₂₀₀ had the lowest levels (0.71–0.74 μ molg^{−1} FW), indicating decreased lipid peroxidation (Table 5). The SOD activity

Table 4

Effects of vermicompost-duckweed (VC-DW) treatments on rice yield, grain filling, and 1000-grain weight. Different lowercase letters indicate statistically significant differences at $p < 0.05$. Data are presented as mean $\pm$ standard error ($n = 3$).

Treatments	Grain yield g pot ⁻¹	Number of filled grains panicle ⁻¹	Number of unfilled grains panicle ⁻¹	1000-grain weight g
Control	22.8 $\pm$ 0.79 ^c	65.9 $\pm$ 1.59 ^a	9.7 $\pm$ 0.16 ^a	20.5 $\pm$ 0.41 ^b
VC_{1.6}DW₆₀	24.76 $\pm$ 0.36 ^{bc}	69.96 $\pm$ 1.6 ^a	8.9 $\pm$ 0.18 ^{ab}	22.19 $\pm$ 0.71 ^{ab}
VC_{4.4}DW₆₀	27.96 $\pm$ 0.25 ^{ab}	73.17 $\pm$ 1.41 ^a	8.1 $\pm$ 0.14 ^{ab}	24.13 $\pm$ 0.71 ^{ab}
VC_{1.6}DW₃₄₀	26.41 $\pm$ 0.59 ^{abc}	71.45 $\pm$ 1.9 ^a	8.6 $\pm$ 0.11 ^{ab}	22.68 $\pm$ 0.65 ^{ab}
VC_{4.4}DW₃₄₀	29.73 $\pm$ 0.73 ^{ab}	74.87 $\pm$ 1.84 ^a	7.3 $\pm$ 0.06 ^{ab}	25.82 $\pm$ 0.37 ^a
VC₁DW₂₀₀	25.11 $\pm$ 0.52 ^{bc}	70.33 $\pm$ 1.36 ^a	8.2 $\pm$ 0.12 ^{ab}	22.73 $\pm$ 0.4 ^{ab}
VC₅DW₂₀₀	30.12 $\pm$ 0.45 ^a	75.38 $\pm$ 0.44 ^a	7 $\pm$ 0.07 ^b	26.29 $\pm$ 0.64 ^a
VC₃DW₀	26.37 $\pm$ 0.64 ^{abc}	72.05 $\pm$ 0.37 ^a	8.9 $\pm$ 0.13 ^{ab}	22.77 $\pm$ 0.57 ^{ab}
VC₃DW₄₀₀	27.51 $\pm$ 0.37 ^{abc}	73.47 $\pm$ 1.63 ^a	8 $\pm$ 0.11 ^{ab}	24.59 $\pm$ 0.42 ^{ab}
VC₃DW₂₀₀	28.42 $\pm$ 0.76 ^{ab}	74.26 $\pm$ 1.8 ^a	7.7 $\pm$ 0.19 ^{ab}	25.45 $\pm$ 0.69 ^a
VC₃DW₂₀₀	28.73 $\pm$ 0.54 ^{ab}	74.39 $\pm$ 1.23 ^a	7.6 $\pm$ 0.2 ^{ab}	25.34 $\pm$ 0.22 ^a
VC₃DW₂₀₀	28.13 $\pm$ 0.24 ^{ab}	75.08 $\pm$ 1.23 ^a	7.7 $\pm$ 0.09 ^{ab}	25.29 $\pm$ 0.61 ^a

was considerably lower in all VC-DW treated treatments in comparison to the control (145 U g⁻¹ FW). The minimum in SOD activity was in VC₅DW₂₀₀ (87.8 U g⁻¹ FW), demonstrating less ROS accumulation (Table 5). The CAT activity was also significantly dropped from 81.0 (U min⁻¹ g⁻¹ FW) in the control to 53.7–57.5 (U min⁻¹ g⁻¹ FW) in VC₃DW₂₀₀ treatments (Table 5).

The impact of VC and DW on the growth responses of rice can be evaluated by analyzing the absolute value of the regression coefficient in the equation (Table 6). Positive and negative coefficients indicate synergistic or antagonistic effects, respectively. The coefficients β_1 and β_2 were positive, indicating that VC and DW additions had a synergistic impact on various growth traits of rice. The absolute value of the coefficient β_{12} showed the coupling effects of VC $\times$ DW on different growth traits of rice (Table 6).

3.1. Principal component analysis and heatmap

Principal component analysis (PCA) and heatmap analysis were conducted further to investigate the relationship between different growth parameters of rice. The first two principal components explained 92.32 % of the total variation, with PC1 being positively correlated with below-ground biomass (BGB), plant height (PH), number of filled grains (NFG), above-ground biomass (ABG), 1000 grain weight (1000-GW), grain yield g pot⁻¹ (GY), root length (RL), SPAD value and panicle length (PL), and negatively correlated with Cd content, MDA content, activities of SOD, CAT and number of unfilled grains (NUG) (Fig. 1a). The heatmap analysis showed two main clusters, where treatments VC₅DW₂₀₀, VC_{4.4}DW₃₄₀, VC₃DW₂₀₀ and VC₃DW₄₀₀ were clustered in the left direction. These treatments are responsible for low levels of Cd in rice grain, lower MDA content, NUG, SOD and CAT activities. In contrast, control treatment along with VC_{1.6}DW₆₀, VC₁DW₂₀₀, VC₃DW₀, and VC_{1.6}DW₃₄₀ were clustered in the right direction of the heatmap plot and produced high levels of the above-mentioned growth attributes and low levels of PL, PH, BGB, NFG, GY, 1000-GW, SPAD, RL, and AGB (Fig. 1b).

Table 5

The impact of several vermicompost-duckweed (VC-DW) treatments on above-ground biomass, below-ground biomass, cadmium (Cd) concentration in rice grains, malondialdehyde (MDA) levels, superoxide dismutase (SOD) activity, and catalase (CAT) activity. Different lowercase letters indicate statistically significant differences at $p < 0.05$. Data are presented as mean $\pm$ standard error ($n = 3$).

Treatments	Above ground biomass g	Below ground biomass G	Grain Cd mg kg ⁻¹	MDA μ mol g ⁻¹ FW	SOD U g ⁻¹ FW	CAT U min ⁻¹ g ⁻¹ FW
Control	34.9 $\pm$ 0.28 ^c	15.7 $\pm$ 0.59 ^d	0.35 $\pm$ 0.003 ^a	2.12 $\pm$ 0.1 ^a	145 $\pm$ 1.33 ^a	81 $\pm$ 4.05 ^a
VC_{1.6}DW₆₀	38.46 $\pm$ 0.72 ^{de}	20.89 $\pm$ 0.78 ^{bc}	0.18 $\pm$ 0.003 ^b	1.33 $\pm$ 0.04 ^b	128.2 $\pm$ 1.03 ^b	74.7 $\pm$ 5.47 ^{ab}
VC_{4.4}DW₆₀	41.23 $\pm$ 0.94 ^{abcd}	24.69 $\pm$ 0.43 ^a	0.15 $\pm$ 0.001 ^{cd}	0.82 $\pm$ 0 ^d	112.5 $\pm$ 0.88 ^c	71.4 $\pm$ 3.09 ^b
VC_{1.6}DW₃₄₀	40.11 $\pm$ 0.37 ^{bcd}	19.27 $\pm$ 0.47 ^c	0.13 $\pm$ 0.002 ^{ef}	1.09 $\pm$ 0.09 ^c	119.2 $\pm$ 8.53 ^{bc}	68.4 $\pm$ 3.03 ^{bc}
VC_{4.4}DW₃₄₀	43.86 $\pm$ 0.76 ^{ab}	26.48 $\pm$ 0.83 ^a	0.07 $\pm$ 0.001 ⁱ	0.75 $\pm$ 0.01 ^d	90.6 $\pm$ 1.75 ^d	57.1 $\pm$ 3.61 ^d
VC₁DW₂₀₀	39.49 $\pm$ 0.31 ^{cd}	18.46 $\pm$ 0.53 ^{cd}	0.16 $\pm$ 0 ^c	1.26 $\pm$ 0 ^b	120.4 $\pm$ 2.09 ^{bc}	73.2 $\pm$ 3.8 ^{ab}
VC₅DW₂₀₀	43.27 $\pm$ 0.75 ^{abc}	25.83 $\pm$ 0.7 ^a	0.08 $\pm$ 0.001 ^{hi}	0.77 $\pm$ 0.01 ^d	87.8 $\pm$ 6.96 ^d	60.1 $\pm$ 1.32 ^{cd}
VC₃DW₀	38.73 $\pm$ 0.35 ^{de}	23.96 $\pm$ 0.2 ^{ab}	0.14 $\pm$ 0.004 ^{de}	0.99 $\pm$ 0.02 ^c	130 $\pm$ 0.93 ^b	75.8 $\pm$ 1.64 ^{ab}
VC₃DW₄₀₀	42.24 $\pm$ 0.12 ^{abcd}	26.51 $\pm$ 0.64 ^a	0.09 $\pm$ 0.001 ^h	0.8 $\pm$ 0.01 ^d	97.2 $\pm$ 4.77 ^d	57.5 $\pm$ 0.21 ^d
VC₃DW₂₀₀	44.52 $\pm$ 0.8 ^a	25.6 $\pm$ 0.21 ^a	0.12 $\pm$ 0.003 ^{fg}	0.72 $\pm$ 0.02 ^d	91.5 $\pm$ 5.62 ^d	53.7 $\pm$ 0.31 ^d
VC₃DW₂₀₀	43.97 $\pm$ 1.21 ^{ab}	26 $\pm$ 0.49 ^a	0.11 $\pm$ 0.004 ^g	0.74 $\pm$ 0 ^d	89.3 $\pm$ 4.17 ^d	55.1 $\pm$ 0.5 ^d
VC₃DW₂₀₀	44.32 $\pm$ 0.64 ^a	25.3 $\pm$ 0.36 ^a	0.12 $\pm$ 0.002 ^{fg}	0.71 $\pm$ 0.01 ^d	87.9 $\pm$ 4.12 ^d	54.2 $\pm$ 0.15 ^d

3.2. Fitting the RSM and identification of optimum VC and DW combination

The ideal combination of VC and DW application doses was determined using RSM to achieve the highest yield and the lowest Cd content in the rice grain. Based on the analysis, the best solution was found to be VC (4.4 %) and DW (187.0 g m⁻²), which resulted in higher GY (29.64 g pot⁻¹) and lower Cd content (0.099 ppm) of rice (Fig. 2).

4. Discussion

With challenges of climate change, soil degradation, environmental toxicity, and food insecurity, there is an increasing need for sustainable and eco-friendly soil management measures (Nguyen et al., 2023). The DW and VC, as an amendment to soil for rice cultivation, is one of such strategies for getting traction. This strategy has the great potential to fulfil the need for environmentally friendly and sustainable soil management practices, as well as to correct heavy metal spoiled soils.

In the present study, we focused on the effect of VC and DW on rice growth, yield, and Cd accumulation in rice grains. The moderate application of VC and DW significantly enhanced several growth parameters of rice plants, such as plant height, root length, panicle length, SPAD value, above- and below-ground biomass. Remarkably, DW influenced panicle length more significantly, whereas VC exerted its primary effects on rice grain yield. Excitingly, the mutual use of VC and

Table 6

Regression coefficients for the model $Y = \beta_0 + \beta_1A + \beta_2B + \beta_{12}AB + \beta_{11}A^2 + \beta_{22}B^2$ were estimated for multiple response variables, including plant height (PH), root length (RL), panicle length (PL), SPAD value, grain yield per pot (GY), number of filled grains per panicle (NFG), number of unfilled grains per panicle (NUG), 1000-grain weight (1000-GW), above-ground biomass (AGB), below-ground biomass (BGB), cadmium concentration in rice grain (Cd), malondialdehyde (MDA) content, and activities of superoxide dismutase (SOD) and catalase (CAT).

Y	β_0	β_1A	β_2B	$\beta_{12}AB$	$\beta_{11}A^2$	$\beta_{22}B^2$
PH	140.77	2.47***	1.99**	-1.50*	-5.05***	-5.15***
RL	43.03	1.37**	1.35**	0.76ns	-1.60**	-1.11**
PL	18.73	0.55**	1.18***			
SPAD	47.43	1.56***	1.39***	0.26ns	-0.66*	-0.50*
GY	28.43	1.70***	0.63**	0.03ns	-0.42*	-0.76**
NFG	74.58	1.72***	0.65**	0.05ns	-0.97**	-1.02**
NUG	7.63	-0.45***	-0.31**	-0.12ns	0.02ns	0.47***
1000-GW	25.36	1.26***	0.59**	0.30ns	-0.52**	-0.94***
AGB	44.27	1.48***	1.16***	0.25ns	-1.45***	-1.90***
BGB	25.63	2.68***	0.47ns	0.85ns	-1.96**	-0.41ns
Cd	0.12	-0.03***	-0.03***			
MDA	0.72	-0.19***	-0.07**	0.04ns	0.16***	0.10**
SOD	89.57	-11.30***	-9.66***	-3.23ns	8.21**	12.96***
CAT	54.33	-4.14***	-5.81***	-2.00*	6.47***	6.47***

ns stands for not significant.

* $P < 0.05$.

** $P < 0.01$.

*** $P < 0.001$ denote the significance of effects from ANOVA.

DW significantly improved the number of filled grains and decreased the number of unfilled grains, signifying the positive relationship of these amendments.

The properties of VC and DW may be responsible for their positive effect on rice growth and production (Oyege and Balaji Bhaskar, 2023). The VC was rich with nitrogen, phosphorus, and potassium, which enhanced soil fertility and facilitated plant growth (Teka et al., 2024). As supported by earlier studies performed by Qasim et al. (2023), where it was shown that the VC can improve physical, chemical, and biological properties of soil, hence increase growth parameters of rice, including plant height, root length, panicle length, and SPAD value. VC increases soil health and plant nutrient availability due to the presence of beneficial microorganisms (Pereira et al., 2023). In our previous study, we also observed that VC enhanced biomass and grain yield of rice in soil contaminated with heavy metals (Roy et al., 2024a). Our study presents new understandings of the interaction effects between VC and DW on Cd accumulation in rice grains. Soil contaminated with heavy metals is of great concern to agricultural productivity, crop quality and ultimately to human health (Rashid et al., 2023; Roy et al., 2024b). Our study results indicated significant reductions in Cd content in rice grains with VC or DW application, but were most promising when used together. This finding aligns with but also extends previous studies by demonstrating the enhanced effectiveness of the combined amendments in reducing Cd content.

The phytoremediation properties of DW allowed it to sequester Cd from the soil, reducing its availability to plants, and several recent studies showed similar impacts (Roy et al., 2024a; Wang et al., 2022; Zhou et al., 2023). The addition of DW to Cd-contaminated soil can also increase several growth parameters of rice seedlings (Hossain et al., 2022; Weerakkodi et al., 2020). Similarly, VC reduced the availability of Cd in soil by binding with Cd ions, immobilizing them, and improving soil health and nutrient availability, as supported by previous studies (Alvarenga et al., 2022; Kibria et al., 2023). Numerous investigations have indicated that organic amendments significantly diminish Cd availability in soil and mitigate Cd buildup in rice grains (Liu et al., 2020, 2021).

Furthermore, exposure to Cd stress induces oxidative damage in rice seedlings, as evidenced by the excessive buildup of MDA content under stress conditions (Chen et al., 2024). Fortunately, the use of VC and DW leads to a significant augmentation in the activities of antioxidant enzymes such as SOD and CAT, ultimately resulting in a reduction in MDA content. Since VC is rich in essential macro-nutrients and provides plants

with significant amounts of micronutrients, it functions as a prosthetic group for antioxidant enzymes. These enzymes play a crucial role in decomposing reactive oxygen species (ROS) and preventing the degradation of chlorophyll molecules (Dumanović et al., 2021). The SOD is a crucial enzyme responsible for the degradation of superoxide anions, resulting in the generation of H_2O_2 and oxygen (Andrés et al., 2023). To safeguard cells from potential H_2O_2 -induced harm, various protective mechanisms are employed. These include the action of CAT and APX, and several peroxidases, which facilitate the breakdown of H_2O_2 into water (H_2O) and oxygen (O_2) through catalytic processes (Afkari, 2018). The levels of SOD, CAT, and APX can be increased in response to the application of VC. For instance, Afkari (2018) observed greater activity levels of these enzymes in *Borage officinalis*, and Kiran (2019) also reported similar findings in lettuce (*Lactuca sativa* var. *crispa*).

The synergy of VC and DW seems to be based on their different but complementary roles. The VC contains a wealth of elements (N, P, K) essential for rice seedlings and some beneficial soil microorganisms. They enhance soil fertility, structure, and microbial biomass, and immobilize Cd through complexation, adsorption, and redox reactions (Alvarenga et al., 2022; Teka et al., 2024). VC also helps to increase the activity of antioxidant enzymes in the plant and thus can also improve the plant tolerance to Cd-induced oxidative stress (Roy et al., 2024a). DW uptakes and immobilizes bio-available Cd in the soil solution, decreasing the Cd availability to the rice roots (Chen et al., 2023; Zhou et al., 2023). DW also affects rhizosphere ecophysiology through changes in soil pH, nitrogen forms, and microbial communities (Hossain et al., 2022). The synergistic effect might be attributed to VC immobilizing Cd as a solid phase, while DW decreases the bioavailable Cd concentration in the soil solution, hence inhibiting Cd mobility by both chemical stabilization and biological uptake. In addition, VC has a positive effect on the soil microbial community and nutritional status (Tian et al., 2024), which may improve the phytoremediation efficiency of DW by promoting growth and Cd accumulation ability. In the meantime, DW may indirectly promote the VC-mediated microbial processes by encouraging root exudation and microbial diversity.

Our results validated that this combined approach [optimal doses of VC (4.4 %) and DW (187.0 g m⁻²)] not only reduced Cd level in grains [0.099 mg kg⁻¹, level falls below the maximum allowable limit for Cd in rice (Kibria et al., 2022)] but also strikingly increased grain productivity (29.64 g pot⁻¹), SPAD, antioxidant system (SOD, CAT APX), and decreased MDA level. Thus, the synergy we observed is not just additive, but arises from interacting biological and chemical processes that

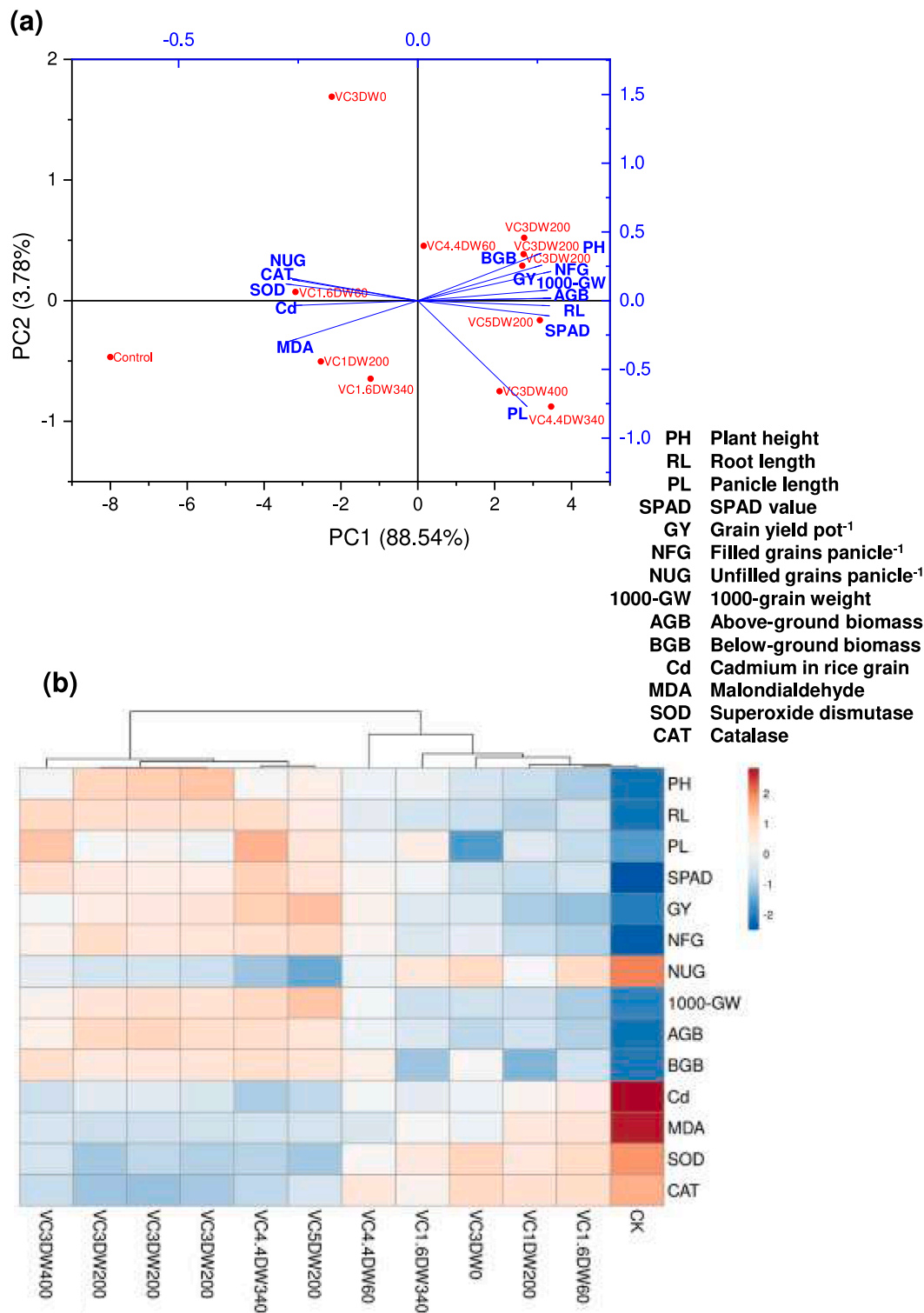
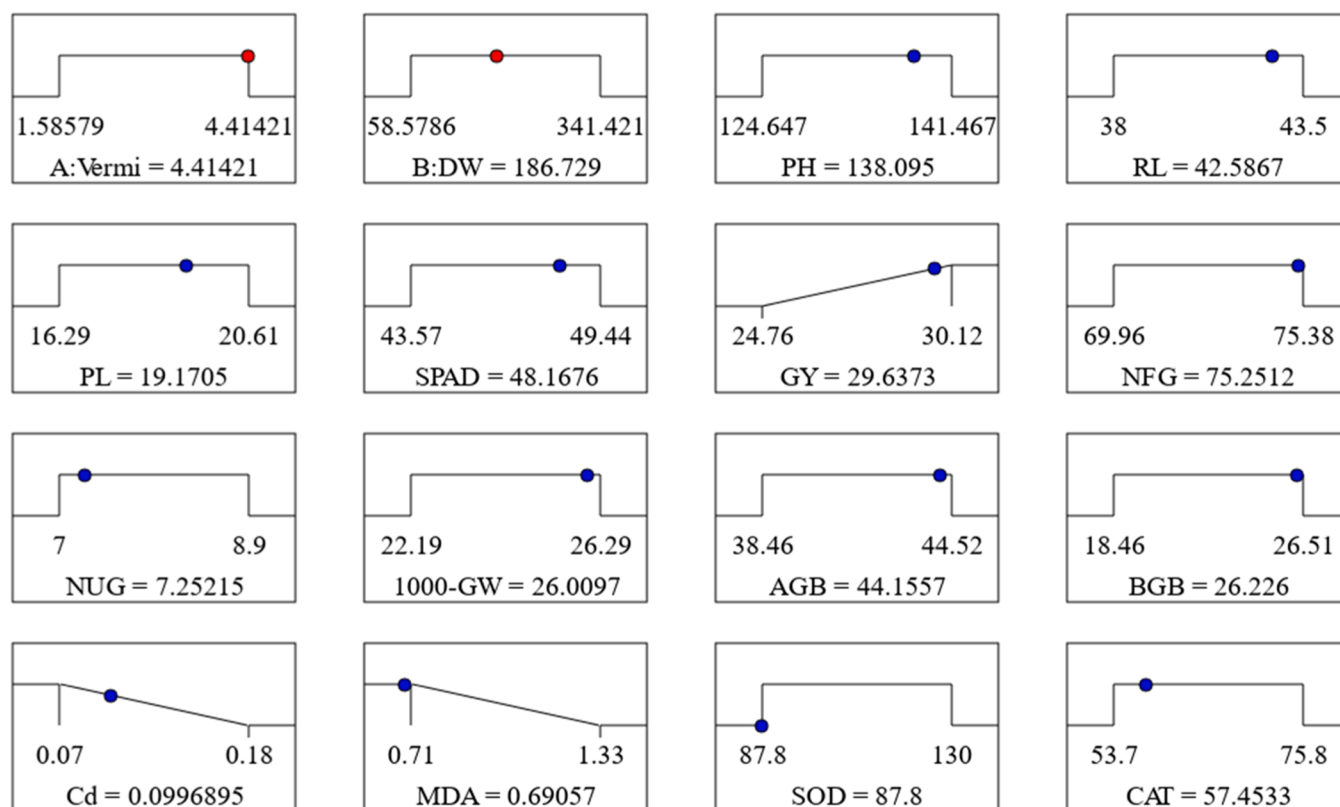


Fig. 1. (a) Principal component analysis and (b) Heatmap reveal the effects of various regimes of vermicompost-duckweed (VC-DW) on the morphological, physiological, and biochemical growth responses of rice.

combine to form a more stable, biologically more enriched and chemically less toxic rhizosphere environment for rice. This integrated strategy is promising for global use in the contaminated paddy and promotes the concept of multi-component strategy-based design for heavy metal remediation.

4.1. Comparisons and original contributions of the current study

This research proposes an innovative approach to Cd remediation in rice crop by combining VC and DW as organic amendment and phytoremediation, respectively. The protective effect of VC on plant Cd tolerance has been reported in previous studies. For instance in their various studies (Iqbal et al. 2024a, 2024b, 2024c) reported that VC



Desirability = 0.873
Solution 1 out of 2

Fig. 2. A desirability ramp illustrates the optimization of duckweed (DW) and vermicompost (VC).

increased leaf physiological processes and antioxidant enzymes activities and improved photosynthetic performance, leading to a decrease in Cd level and an increase in yield. Other organic amendments, including compost and manure, have also achieved good results. It is reported that composts could decrease the bioavailability of Cd via the combined effects of microbial activity and chemical processes, for example complexation and redox reactions (Gutiérrez et al., 2022). Bashir et al. (2020) also observed that grain Cd concentrations decreased significantly, and yield increased with co-composted biochar and farmyard manure. Furthermore, in previous work (Roy et al., 2024a), we found that the combined addition of biochar, VC, and DW alleviated arsenic-induced stress in rice, suggesting that a combined organic addition remediation strategy may be beneficial. However, Shen et al. (2022) reported an increase in Cd uptake with VC, and which suggested the necessity of site-specific evaluation of amendment strategies. Regarding phytoremediation, DW has demonstrated promise for Cd phytoremediation. Chen et al. (2023) found that DW could reduce Cd contents in rice grains by changing soil pH, nitrogen species and microbial communities. Additionally, Zhao et al. (2024) showed that the application of ramie (*Boehmeria nivea* L.) by means of floating island technology was effective for in situ field-scale Cd remediation.

The current study distinguishes itself by introducing a novel, synergistic strategy through the combined use of VC and DW, which has not been previously tested for Cd remediation in rice systems. This integrated approach resulted in improved plant physiological performance, strengthened antioxidative responses, reduced oxidative stress, and minimized Cd translocation to edible plant parts. In contrast to earlier studies that evaluated amendments or phytoremediation agents in isolation, the present research offers a comprehensive, sustainable, and field-applicable solution. Our findings contribute to the development of

long-term, eco-friendly management practices for mitigating heavy metal pollution in paddy soils, with implications for safe and productive rice cultivation.

4.2. Fate of Cd in the post-harvest and its management

Although our present research primarily focuses on the reduction in Cd uptake in rice grains by VC and DW amendments, the fate of remaining Cd in soil and other plant portions after the harvest is also worth discussing. Following rice cultivation, Cd can remain in the soil matrix and be mobilised by leaching or runoff. Moreover, rice residues (straw, husks, roots) can also accumulate Cd, and if re-entered in the field as a means of compost, mulch can again contribute bioavailable Cd if not adequately treated. Thus, some innovative methods, such as adopting phytoremediation approaches using hyperaccumulators during the intermediate period of crop rotation (Yang et al., 2022), the use of engineered microbial consortium for Cd sequestration (Che et al., 2024), and Cd recovery by chelate-assisted extraction (Dong et al., 2023), provide alternative pathways for recycling or removal of Cd. The implementation of these approaches will lead to a more comprehensive and circular approach for the management of Cd and also other heavy metals in agricultural systems and would secure the long-term sustainability of remediation practices.

4.3. Limitations

Although the results of our pot experiment provided useful information, the effect of some environmental and agronomic parameters on the Cd bioavailability, as well as the ability of added VC and DW to amend soil, might differ under actual field conditions. Soil

heterogeneity, such as soil texture, structure, pH, and CEC, can influence the capabilities of added amendments and also can further influence the mobility and adsorption of Cd (Yu et al., 2023). Moreover, the differences in soil microbial communities, enzyme activities and rhizosphere interactions are important factors for organic amendments transformation and immobilization of Cd (Liu et al., 2024), which can influence our results. Furthermore, agricultural practices (especially water management, waterlogging and drought) can influence the redox status of soil and consequently the Cd speciation and plant availability (Stafford et al., 2018). Even short-term changes in weather conditions (like fluctuations of temperature, precipitation and light intensity) also affect decomposition turnover times of amendments, plant physiology, and the efficacy of remediation. Rice genotypes, root architecture and membrane transport system differences among genotypes also affect Cd uptake and the plant response to soil amendments (Shan et al., 2024). Finally, our single-time pot experiment may not adequately mimic the complexity of field ecosystems (including long-term nutrient dynamics, amendment persistence and seasonality). So, in order to verify the utility of VC and DW under real conditions, several long-term multi-season field experiments in different agro-ecological zones are needed.

5. Conclusions

The utilization of VC and DW in agricultural soils offers significant advantages for enhancing rice growth while avoiding Cd contamination. These supplements improve plant height, panicle length, SPAD value, and grain yield per pot. The synergistic application of VC and DW increases antioxidant enzyme activities, reduces MDA levels, and lowers Cd content in rice grains. RSM analysis indicates that combining VC (4.4 %) and DW (187.0 g m⁻²) maximizes rice yield and minimizes Cd content in the grain. While further field studies need to be carried out for wide-scale validation, we argue that the usage of such a resource combination (VC = 4.4 % and DW = 187.0 g m⁻²) can be an effective and recommended approach for rice cultivation on Cd-contaminated soils.

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Rana Roy: Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Akram Hossain:** Writing – review & editing, Visualization, Investigation. **Tanwne Sarker:** Writing – review & editing, Visualization, Software, Formal analysis. **Muhammad Asaduzzaman:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

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Data availability

Data used in this study are included and the raw data will be provided upon request.

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